7 ethernet

Problem 7.1 — Ethernet Frame Analysis

An Ethernet frame is captured with the following field values (hex):

```
Dst MAC: FF:FF:FF:FF:FF:FF   Src MAC: A4:C3:F0:85:AC:2D
      Type/Length:  0x0806   Payload:  (46 bytes)
```

(a) Is this a unicast, multicast, or broadcast frame? How can you tell? → broadcast because dst address is FF:....

(b) What protocol does the EtherType `0x0806` indicate? → ARP

(c) What is the total frame size in bytes (excluding preamble and SFD)? → 6 + 6 + 2 + 46 + 4 = 64 bytes

(d) What is the purpose of the 7-byte preamble and 1-byte SFD? → 7 byte preamble: allow sync rate to match up; SFD: signal start & end of data

(e) The FCS covers which fields of the frame? → all but preamble / SFD

DST, SRC, Type / len, payload

Problem 7.2 — ARP Trace

Host A (IP: `10.0.0.1`, MAC: `AA:AA:AA:AA:AA:AA`) wants to send an IP packet to Host B (IP: `10.0.0.2`, MAC: `BB:BB:BB:BB:BB:BB`). Both hosts are on the same subnet. Host A has an empty ARP cache. → nothing saved from IP address to MAC

ARP: I am... looking for...

(a) Describe the ARP request that A sends. What is the destination MAC address of the Ethernet frame carrying this request, and why? → sends looking for IP of B ↳ sends to all on the LAN to try to find right device; FF:FF:FF: broadcasted

(b) Host B receives the request. Describe the ARP reply it sends. → I am .... unicast back

(c) After the ARP exchange, Host A sends the IP packet. Fill in the fields of the Ethernet frame:

- Source MAC: AA:AA:
- Destination MAC: BB:BB ...
- EtherType: IPv4

(d) Host A now wants to send a packet to Host C (IP: `192.168.5.1`) on a *different* subnet. The default gateway has IP `10.0.0.254` and MAC `GG:GG:GG:GG:GG:GG`. What destination MAC address does A use in the Ethernet frame?

↳ send to router → GG:GG:

Problem 7.3 — Switch Self-Learning

Consider the network below. Switch S1 has an empty forwarding table. Three hosts are connected: Host A on port 1, Host B on port 2, Host C on port 3.

(a) A sends a frame to B. What does S1 do? What does S1's forwarding table contain after this? → S1 → A in port 1 ↳ send message to port 2 & 3 (floods)

(b) B sends a frame to A. What does S1 do? What does S1's forwarding table contain after this? → S1 → A in port 1, B in port 2 ↳ send to port 1

(c) A sends a frame to C. What does S1 do? → ~~send to port 3~~

not in table so floods port 2 & 3 even though we know B is in 2 we don't know where C is

(d) What would happen if there were a *loop* in the switch topology (e.g., a second cable connecting the switch back to itself)? How is this problem solved in practice?

↳ forever loop → broadcast storm ↳ spanning tree protocol (STP)

Problem 7.4 — Ethernet Generations

(a) Match each Ethernet standard to its line code:

Standard	Line code
10BASE-T	? manchester
100BASE-TX	? 4B5B & MLT
1000BASE-T	? 4D PAM

Choices: Manchester encoding; 4B5B + MLT-3; 4D-PAM5

(b) What does "BASE" signify in the naming convention? →

(c) Why does 1000BASE-T require *four* wire pairs, while 100BASE-TX uses only two?